

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |       |  | Nerve net (1)<br>Unmyelinated / branched/ bidirectional impulses / impulses travel in many directions (1)                                                                                                                                                                                                                                   | 2               |     |     | 2     |       |      |
|          | (b) | (i)   |  | Synaptic {knob / bulb} / axon terminal                                                                                                                                                                                                                                                                                                      | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | Vesicle: synaptic<br>Accept secretory / excretory / neurosecretory<br><b>and</b><br>Role: transports {neurotransmitter / acetyl choline}<br><b>Name and function for 1 mark</b>                                                                                                                                                             | 1               |     |     | 1     |       |      |
|          |     | (iii) |  | {Acetylcholine / Neurotransmitter} {not broken down / remains in synaptic cleft} (1)<br><br>Continues to stimulate {post synaptic membrane / adjacent cell}/<br>Continuous {depolarisation / action potentials} in {adjacent cell / neurone} /<br>more {Acetylcholine / Neurotransmitter} binding to receptor on post-synaptic membrane (1) |                 | 2   |     | 2     |       |      |
|          | (c) | (i)   |  | X- <b>depolarisation</b><br>Y- <b>repolarisation</b><br><b>both for 1 mark</b>                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | (Na <sup>+</sup> ) Channel opens (1)<br>{Na <sup>+</sup> / sodium ions} rush {into axon / into neurone / through membrane} (1)<br>Accept flood/ rapidly diffuse in                                                                                                                                                                          | 2               |     |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (d) |  |  | <b>Any four (×1) from:</b><br>A. Increases (up to a maximum / 60) and then decreases (1)<br>B. Depolarisation only takes place at the nodes / Correct reference to saltatory conduction (1)<br>C. Less membrane needs to be depolarised / action potential does not have to travel the whole length of the neurone (1)<br>D. (The Longer the internode the conduction speed increases) due to {less 'jumps' from node to node / influences of local currents increase} (1)<br>E. When internodes are too long, {impulses not able to jump from node to node / local currents are unable to extend to next node / no saltatory conduction} (1) |                 | 2   | 2   | 4     |       | 4    |
|          |     |  |  | Question 4 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 7               | 4   | 2   | 13    | 0     | 4    |